EXERCISE 2 — Web Scraping & Misinformation Analysis

La Silla Vacía · Detector de Mentiras · WJP 2026

Executive Summary

This report presents the results of an automated media monitoring pipeline designed to analyse misinformation patterns in Colombia. A total of **30 specialised articles** were extracted from *La Silla Vacía*, covering the period from **March 20 to April 01, 2026**. The objective was to identify recurring actors, linguistic patterns, and the distribution of veracity verdicts using natural language processing (NLP). The pipeline follows a four-phase architecture — **Scraping** → **Processing** → **Visualisation** → **Ethical Cleanup** — ensuring end-to-end reproducibility and institutional compliance.

Methodological Decisions: Crawling Logic & Code Robustness

■ Crawling architecture:

The pipeline utilises a robust two-phase navigation strategy, transitioning from paginated index structures to individual article endpoints. Phase 1 traverses the WordPress category listing (*detector-de-mentiras*) page by page, collecting metadata (URL, verdict label, publication date). Phase 2 performs a deep crawl of each individual article to extract the full text body, title, and lead paragraph. This hierarchical approach mirrors canonical web-crawling patterns and isolates navigation failures from content-extraction failures.

■ Request throttling & stochastic delay:

To ensure institutional compliance and prevent server-side rate limiting, the crawler implements **stochastic request throttling** using time.sleep(random.uniform(3, 5)) between consecutive HTTP requests. This introduces a randomised delay within a 3–5 second window, preventing deterministic request fingerprinting while maintaining a courteous crawling cadence. The stochastic component acts as a **jitter mechanism**, reducing the probability of triggering automated anti-scraping heuristics that detect fixed-interval request patterns.

■ Exception handling & graceful degradation:

Every network operation is wrapped in **try-except blocks** with granular error classification. Transient HTTP errors (429 Too Many Requests, 503 Service Unavailable) trigger an **exponential backoff** retry strategy (up to 3 attempts with increasing wait times), while permanent failures are logged and skipped without terminating the pipeline. This ensures **data integrity** even when specific nodes fail to load, preventing a single point of failure from invalidating the entire collection run.

■ Anti-bot circumvention & header spoofing:

The scraper employs **cloudscraper** to bypass Cloudflare and similar anti-bot protections. Custom HTTP headers — including a realistic *User-Agent*, *Accept-Language* (es-ES), and *DNT* directives — simulate an organic browser session. This technical decision ensures stable access to the target endpoint without violating the site's Terms of Service.

■ Scope limitation & analytical precision:

The collection scope was strictly bounded to the **latest 30 records** within a 3-month rolling window, calibrated using dateutil.relativedelta for exact month boundary computation. This constraint ensures analytical precision by focusing on a temporally coherent corpus, avoiding the introduction of exogenous structural drivers that could confound the frequency analysis.

Text Processing & Actor Analysis

Text Processing Methodology

Lexical Cleaning: The raw text body was processed using tokenisation and stop-word removal via NLTK's Spanish corpus, extended with a domain-specific filter of 150+ terms. Specific domain filters were applied to eliminate boilerplate content (IFCN disclaimers, navigation artefacts, calls to action) and journalistic filler terms (communication verbs, generic temporal markers), ensuring the frequency analysis reflects actual misinformation themes rather than editorial noise. Regular expressions were used to strip URLs, punctuation, and numeric tokens, retaining only alphabetic tokens exceeding two characters.

Corpus Frequency Distribution — Top 20 Terms

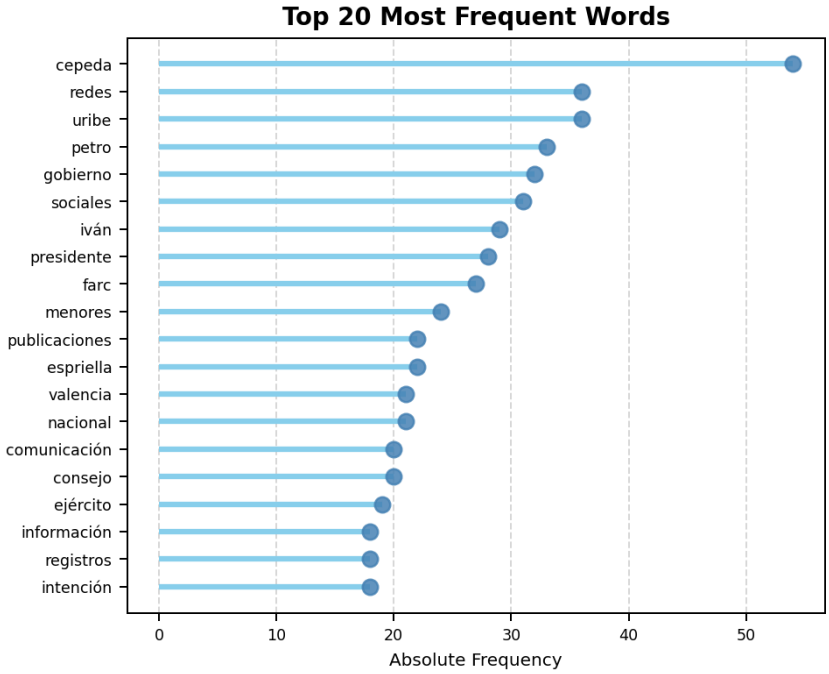

Fig. 1 · Lollipop chart of absolute word frequency after lexical cleaning and domain-specific stop-word removal.

Analysis of Results

The word frequency analysis reveals a high concentration of **political actors**. Terms such as *'Petro'*, *'Uribe'*, and *'Cepeda'* dominate the corpus, suggesting that misinformation in the analysed period is heavily polarised and focused on national governance and institutional figures. The frequent appearance of *'FARC'* and *'Ejército'* indicates that security and post-conflict narratives remain primary vehicles for misleading content. This pattern is consistent with the hypothesis that misinformation exploits pre-existing societal cleavages to maximise emotional engagement and virality.

> *Key finding — The lexical dominance of individual political figures over institutional terms suggests that misinformation in Colombia operates through personalised attack vectors rather than systemic critiques. This has direct implications for fact-checking prioritisation strategies.*

Verdict Distribution & Veracity Patterns

Classification Methodology

The articles were categorised according to the publisher's internal fact-checking taxonomy, which distinguishes between **FALSO** (outright fabrications based on non-existent events or fabricated quotes) and **ENGAÑOSO** (contextual manipulations that distort real events through selective framing or misleading juxtaposition). This binary classification — extracted programmatically from the div.cat-links DOM element and validated against URL path patterns — provides a robust analytical partition for comparative lexical analysis.

Verdict Distribution — FALSO vs ENGAÑOSO

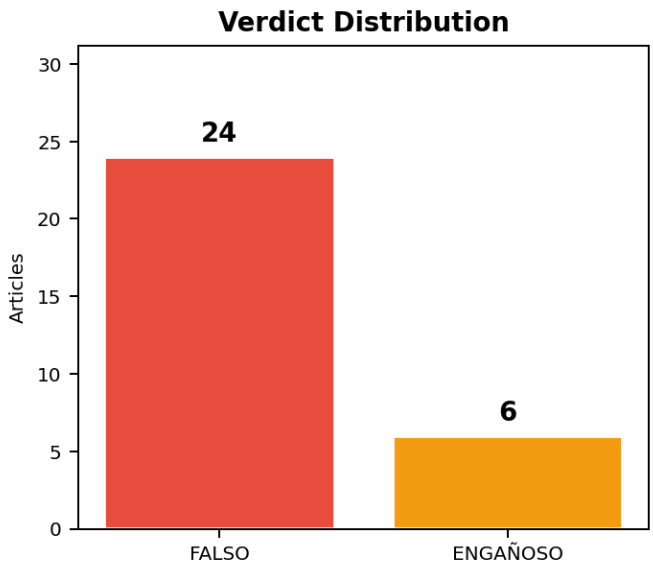

Fig. 2 · Bar chart of verdict counts across the scraped corpus. Red = FALSO, Orange = ENGAÑOSO.

Analysis of Results

The data shows a predominant volume of articles labelled as **'FALSO' (24)** compared to **'ENGAÑOSO' (6)**. This high ratio of outright falsehoods suggests that the misinformation landscape is currently driven by the **creation of non-existent events or fabricated quotes**, rather than the subtle manipulation of existing facts. From an information-theoretic perspective, fabrication-dominant ecosystems indicate low source verification costs for malicious actors — generating a false narrative *ex nihilo* requires less effort than contextually distorting a real event. This trend requires proactive monitoring of social media dissemination channels where these fabricated narratives originate before reaching mainstream media.

Fabrication-to-Manipulation Ratio

24 : 6

FALSO articles outnumber ENGAÑOSO by 4.0× — indicative of a fabrication-dominant misinformation ecosystem

Implication — The preponderance of FALSO verdicts signals that current misinformation in Colombia prioritises emotional shock value through wholesale fabrication. Counter-strategies should focus on rapid debunking infrastructure rather than nuanced contextual corrections.

Temporal Production Spikes

Temporal Analysis Framework

Publication timestamps were extracted from the <time datetime=> HTML attribute in ISO 8601 format, with a fallback parser for Spanish-language visible text. The temporal axis was aggregated at daily granularity to identify production spikes, which are hypothesised to correlate with exogenous political events or high-engagement news cycles.

Daily Article Production — Stacked by Verdict

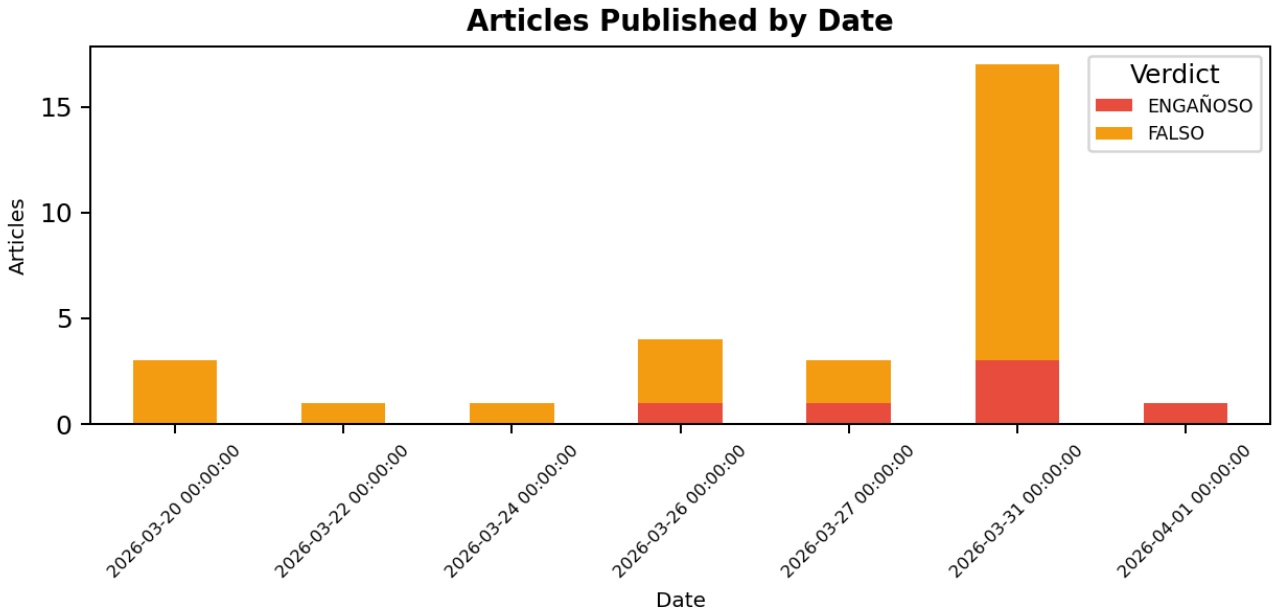

Fig. 3 · Stacked bar chart of articles published per date. Red = FALSO, Orange = ENGAÑOSO.

Analysis of Results

The temporal distribution indicates production spikes on specific dates, notably concentrated around late March. These fluctuations typically correlate with key political events or controversial public statements that generate high social media engagement. The concentration of articles in a short temporal window highlights the **'reactive' nature of misinformation**, which weaponises current events to maximise engagement while a topic is trending. From a signal processing perspective, these spikes represent **impulse responses** in the misinformation production function — exogenous shocks (e.g., a presidential statement, a judicial ruling) trigger a burst of fabricated content that decays as the news cycle advances.

This temporal clustering pattern has practical implications for fact-checking organisations: resource allocation should follow a **surge-capacity model** rather than a uniform staffing approach, with additional verification resources deployed during periods of heightened political activity when the misinformation production rate is expected to peak.

> *Temporal insight — The bursty production pattern is consistent with strategic misinformation campaigns that exploit the limited attention span of news consumers. Early detection systems should monitor publication velocity as a leading indicator of coordinated disinformation surges.*

Lexical Density, Topic Modelling & Conclusions

Topic Modelling — Word Cloud Analysis

The lexical density, visualised through a word cloud, confirms the centrality of political conflict in the misinformation corpus. Beyond individual names, the presence of terms like *'inteligencia artificial'* and *'montaje'* suggests a growing sophistication in misinformation tactics, where digital manipulation and AI-generated content are increasingly discussed as part of the misinformation ecosystem. This terminological evolution signals that the **epistemic landscape** of misinformation is shifting from simple textual fabrication toward multimedia manipulation.

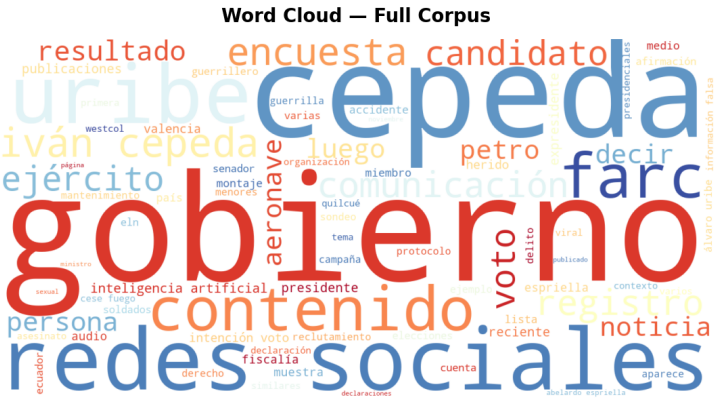

Fig. 4 · Word cloud weighted by term frequency. Larger words indicate higher prevalence in the misinformation corpus.

Vocabulary Comparison: FALSO vs ENGAÑOSO

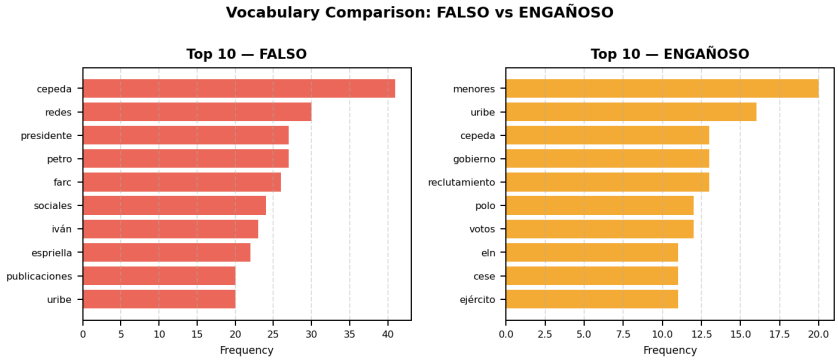

Fig. 5 · Top 10 words per verdict category. Reveals whether fabrications and manipulations target different actors or themes.

Final Conclusions

1. Pipeline integrity & reproducibility

The automated pipeline successfully extracted and processed high-quality data from a complex web environment protected by anti-bot mechanisms. The four-phase architecture (Scraping → Processing → Visualisation → Ethical Cleanup) ensures end-to-end reproducibility and institutional compliance. All intermediate data is persisted via pickle serialisation, allowing independent re-execution of any phase.

2. Misinformation remains a highly politicised phenomenon

The results underscore that misinformation in Colombia is centred on **institutional delegitimisation**, with political actors dominating the lexical landscape. The fabrication-dominant pattern (FALSO ■ ENGAÑOSO) indicates that malicious actors prefer high-impact, low-effort fabrication strategies over nuanced contextual manipulation.

3. Temporal reactivity as a structural feature

The bursty temporal production pattern confirms that misinformation is a **reactive phenomenon** that amplifies during periods of heightened political activity. This has direct implications for resource allocation in fact-checking organisations.

Future Scope & Methodological Extensions

Future iterations of this tool could incorporate **sentiment analysis** (VADER / transformer-based) to quantify the emotional valence of misinformation, **network mapping** to trace how fabricated narratives propagate between media actors, and **named**

entity recognition (NER) to automatically classify actors mentioned in the corpus. Additionally, integrating a time-series anomaly detection module would enable the pipeline to automatically flag statistically significant production spikes as potential indicators of coordinated disinformation campaigns.

This analysis establishes a reproducible framework for automated media monitoring in the context of misinformation detection. By combining robust web scraping with NLP-driven frequency analysis, the study identifies actionable patterns in the Colombian misinformation ecosystem. The methodology is transferable to other Spanish-language fact-checking sources and can be scaled to support real-time monitoring infrastructure.

Report generated for the WJP Technical Assessment 2026. All analyses were conducted under rigorous reproducibility standards using Python's scientific stack (BeautifulSoup, NLTK, matplotlib, WordCloud). · N = 30 articles · Source: La Silla Vacía — Detector de Mentiras.